

Open Source Frameworks for Big Data Processing: An Overview

This article discusses the Big Data processing ecosystem and the associated architectural stack. It investigates different frameworks suiting the various processing requirements of Big Data. It also delves into the frameworks at various layers of the stack such as storage, resource management, data processing, querying and machine learning.



The initial growth of data on the Internet was primarily driven by a greater population gaining access to the Web. This improved access was fuelled by the advent of a newer range of devices like smartphones and tablets. Riding on the first generation of data growth, a second wave of scaled up data production was unleashed mainly by the social media platforms that drove the growth upwards, exponentially. The collaborative nature of the information-sharing platforms contributed to a viral growth of the data shared via these platforms. The third wave of data generation is largely being led by the proliferation of intelligent connected devices and will lead to a scale of data generation that is unprecedented.

In addition, the advancements in scientific fields coupled with the availability of cheaper computing has led to newer applications in the fields of medical sciences, physics, astronomy, genetics, etc, where large volumes of data are collected and processed to validate hypotheses, and enable discoveries and inventions.

The huge growth in data acquisition and storage has led us to the next logical phase, which is to process this vast volume of data to make sense of it. The need to process this vast volume of data has led to the demand for scalable and parallel systems that process the data at speed and scale. Open source technologies are a natural choice for the high performance computing needed for large scale data processing.